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K-MEANS

DISUSUN OLEH:

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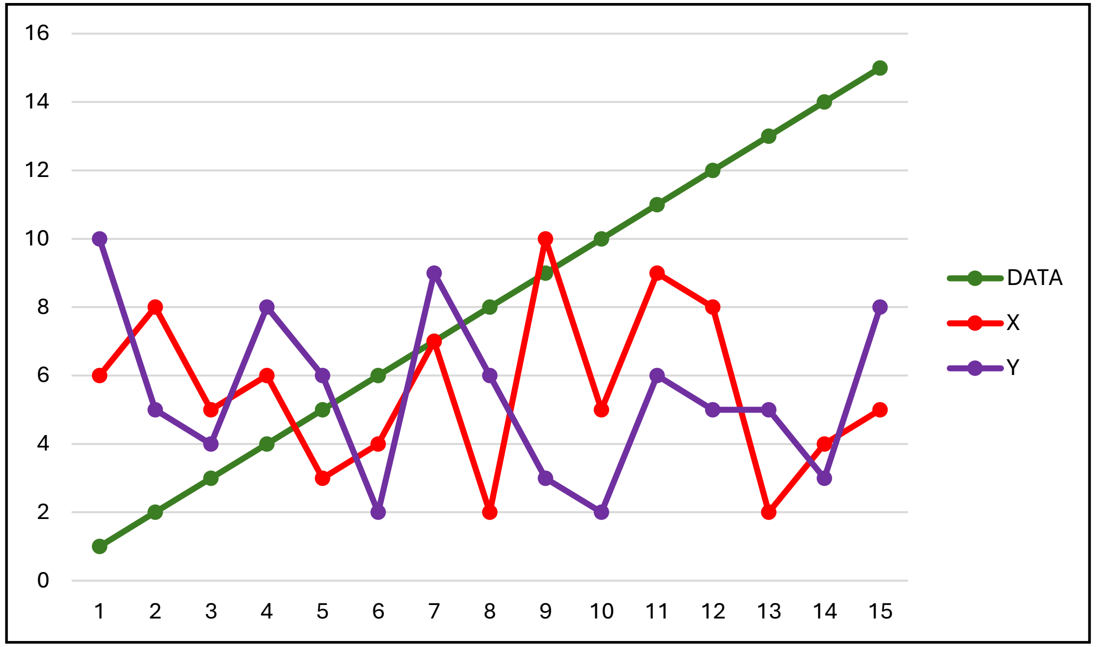
LANGKAH K-MEANS

1. Siapkan dataset
2. Tentukan jumlah cluster (k)
3. Pilih titik centroid secara acak
4. Kelompokkan data sehingga terbentuk (K) buah cluster dengan titik centroid dari setiap cluster
5. Perbarui nilai titik centroid
6. Ulangi Langkah 3 sampai 5 sampai nilai titik centroid tidak lagi berubah

LANGKAH PENYELESAIAN

DATA KE-	X	Y
1	6	10
2	8	5
3	5	4
4	6	8
5	3	6
6	4	2
7	7	9
8	2	6
9	10	3
10	5	2
11	9	6
12	8	5
13	2	5
14	4	3
15	5	8

CENTROID	X	Y	SIMBOL
C1	2	4	
C2	5	8	
C3	9	7	



PENGELOMPOKKAN DATA

CENTROID	X	Y
C1	2	4

DATA	X	Y	C1	HASIL
1	6	10	$\sqrt{(6-2)^2+(10-4)^2}$	7,211102551
2	8	5	...	6,08276253
3	5	4	...	3
4	6	8	...	5,656854249
5	3	6	...	2,236067977
6	4	2	...	2,828427125
7	7	9	...	7,071067812
8	2	6	...	2
9	10	3	...	8,062257748
10	5	2	...	3,605551275
11	9	6	...	7,280109889
12	8	5	...	6,08276253
13	2	5	...	1
14	4	3	...	2,236067977
15	5	8	...	5

PENGELOMPOKKAN DATA

CENTROID	X	Y
C2	5	8

DATA	X	Y	C2	HASIL
1	6	10	$\sqrt{(6-5)^2+(10-8)^2}$	2,236067977
2	8	5	...	4,242640687
3	5	4	...	4
4	6	8	...	1
5	3	6	...	2,828427125
6	4	2	...	6,08276253
7	7	9	...	2,236067977
8	2	6	...	3,605551275
9	10	3	...	7,071067812
10	5	2	...	6
11	9	6	...	4,472135955
12	8	5	...	4,242640687
13	2	5	...	4,242640687
14	4	3	...	5,099019514
15	5	8	...	0

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PENGELOMPOKKAN DATA

CENTROID	X	Y
C3	9	7

DATA	X	Y	C3	HASIL
1	6	10	$\sqrt{(6-9)^2+(10-7)^2}$	4,242640687
2	8	5	…	2,236067977
3	5	4	…	5
4	6	8	…	3,16227766
5	3	6	…	6,08276253
6	4	2	…	7,071067812
7	7	9	…	2,828427125
8	2	6	…	7,071067812
9	10	3	…	4,123105626
10	5	2	…	6,403124237
11	9	6	…	1
12	8	5	…	2,236067977
13	2	5	…	7,280109889
14	4	3	…	6,403124237
15	5	8	…	4,123105626

PENGELOMPOKKAN DATA

DATA	X	Y	C1	C2	C3	CENTROID
1	6	10	7,211102551	2,236067977	4,242640687	2
2	8	5	6,08276253	4,242640687	2,236067977	3
3	5	4	3	4	5	1
4	6	8	5,656854249	1	3,16227766	2
5	3	6	2,236067977	2,828427125	6,08276253	1
6	4	2	2,828427125	6,08276253	7,071067812	1
7	7	9	7,071067812	2,236067977	2,828427125	2
8	2	6	2	3,605551275	7,071067812	1
9	10	3	8,062257748	7,071067812	4,123105626	3
10	5	2	3,605551275	6	6,403124237	1
11	9	6	7,280109889	4,472135955	1	3
12	8	5	6,08276253	4,242640687	2,236067977	3
13	2	5	1	4,242640687	7,280109889	1
14	4	3	2,236067977	5,099019514	6,403124237	1
15	5	8	5	0	4,123105626	2

MEMPERBARUI CENTROID

CENTROID LAMA	X	Y
C1	2	4
C2	5	8
C3	9	7

CENTROID BARU	X	Y
C1	3,571428571	4
C2	6	8,75
C3	8,75	4,75

$$C1X = \frac{1}{7}(5 + 3 + 4 + 2 + 5 + 2 + 4) = 3{,}571428571$$

$$C1Y = \frac{1}{7}(4 + 6 + 2 + 6 + 2 + 5 + 3) = 4$$

$$C2X = \frac{1}{4}(6 + 6 + 7 + 5) = 6$$

$$C2Y = \frac{1}{4}(10 + 8 + 9 + 8) = 8{,}75$$

$$C3X = \frac{1}{4}(8 + 10 + 9 + 8) = 8{,}75$$

$$C3Y = \frac{1}{4}(5 + 3 + 6 + 5) = 4{,}75$$

PENGELOMPOKKAN DATA CENTROID BARU

CENTROID	X	Y
CENTROID BARU 1	3,571428571	4

DATA	X	Y	C1	HASIL
1	6	10	$\sqrt{(6-3{,}571428571)^2+(10-4)^2}$	6,472863291
2	8	5	…	4,540071023
3	5	4	…	1,428571429
4	6	8	…	4,67952553
5	3	6	…	2,080031397
6	4	2	…	2,045403009
7	7	9	…	6,062598621
8	2	6	…	2,543499116
9	10	3	…	6,505884307
10	5	2	…	2,457807219
11	9	6	…	5,785273352
12	8	5	…	4,540071023
13	2	5	…	1,862629259
14	4	3	…	1,087967587
15	5	8	…	4,247448214

PENGELOMPOKKAN DATA CENTROID BARU

CENTROID	X	Y
CENTROID BARU 2	6	8,75

DATA	X	Y	C2	HASIL
1	6	10	$\sqrt{(6-6)^2+(10-8{,}75)^2}$	1,25
2	8	5	...	4,25
3	5	4	...	4,85412196
4	6	8	...	0,75
5	3	6	...	4,069705149
6	4	2	...	7,04006392
7	7	9	...	1,030776406
8	2	6	...	4,85412196
9	10	3	...	7,004462863
10	5	2	...	6,823672032
11	9	6	...	4,069705149
12	8	5	...	4,25
13	2	5	...	5,48292805
14	4	3	...	6,087897831
15	5	8	...	1,25

PENGELOMPOKKAN DATA CENTROID BARU

CENTROID	X	Y
CENTROID BARU 3	8,75	4,75

DATA	X	Y	C3	HASIL
1	6	10	$\sqrt{(6-8{,}75)^2+(10-4{,}75)^2}$	5,926634796
2	8	5	...	0,790569415
3	5	4	...	3,824264635
4	6	8	...	4,257346591
5	3	6	...	5,884301148
6	4	2	...	5,4886246
7	7	9	...	4,596194078
8	2	6	...	6,864765109
9	10	3	...	2,150581317
10	5	2	...	4,650268809
11	9	6	...	1,274754878
12	8	5	...	0,790569415
13	2	5	...	6,754628043
14	4	3	...	5,062114183
15	5	8	...	4,96235831

PENGELOMPOKKAN DATA CENTROID BARU

DATA	X	Y	C1	C2	C3	CENTROID BARU
1	6	10	6,472863291	1,25	5,926634796	2
2	8	5	4,540071023	4,25	0,790569415	3
3	5	4	1,428571429	4,85412196	3,824264635	1
4	6	8	4,67952553	0,75	4,257346591	2
5	3	6	2,080031397	4,069705149	5,884301148	1
6	4	2	2,045403009	7,04006392	5,4886246	1
7	7	9	6,062598621	1,030776406	4,596194078	2
8	2	6	2,543499116	4,85412196	6,864765109	1
9	10	3	6,505884307	7,004462863	2,150581317	3
10	5	2	2,457807219	6,823672032	4,650268809	1
11	9	6	5,785273352	4,069705149	1,274754878	3
12	8	5	4,540071023	4,25	0,790569415	3
13	2	5	1,862629259	5,48292805	6,754628043	1
14	4	3	1,087967587	6,087897831	5,062114183	1
15	5	8	4,247448214	1,25	4,96235831	2

MEMPERBARUI CENTROID

CENTROID LAMA	X	Y
C1	3,571428571	4
C2	6	8,75
C3	8,75	4,75

CENTROID BARU	X	Y
C1	3,571428571	4
C2	6	8,75
C3	8,75	4,75

C1X = $\frac{1}{7}(5+3+4+2+5+2+4) = 3{,}571428571$

C1Y = $\frac{1}{7}(4+6+2+6+2+5+3) = 4$

C2X = $\frac{1}{4}(6+6+7+5) = 6$

C2Y = $\frac{1}{4}(10+8+9+8) = 8{,}75$

C3X = $\frac{1}{4}(8+10+9+8) = 8{,}75$

C3Y = $\frac{1}{4}(5+3+6+5) = 4{,}75$

NILAI CENTROID SUDAH SAMA

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